Civet behavior analysis

Phase 2: trained behavior pilot and automatic CCTV evidence

Walking · Feeding · Unknown

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Project goal and Phase 1 continuity

One civet, four connected rooms, four CCTV cameras

Goal: turn CCTV video into traceable behavior information.

Keep behavior separate from visibility and camera conditions.

Continue the Phase 1 pipeline and structured observations.

Use a pretrained detector and a trained behavior classifier.

Behavior definitions and current support

The target ethogram is broader than the trained pilot

Behavior / condition	Activity group	Current output
Walking / Climbing	Active	Walking pilot / Climb unsupported
Feeding	Active	Feeding pilot
Visible resting / sitting	Inactive	Verified result unavailable
Other visible behavior	Depends on behavior	No trained Other class
Hidden / unclear / unsupported	Unknown	Unknown

Data and source coverage

A completed sampling plan across all discovered unique videos

31 original videos from cameras A01–A04.

109.84 camera-hours include overlapping views of one animal.

1,319 camera windows, each nominally 2 seconds.

348 fused nominal times, including 310 with all four cameras.

Automatic analysis workflow

Animal presence and behavior classification have different jobs

1. Inventory files and decode by presentation timestamps.

2. MegaDetector V5A proposes animal presence.

3. Motion features feed the trained Random Forest.

4. Keep uncertain cases as Unknown and combine camera times.

5. Export source-linked tables, plots and evidence.

What we trained

Nine existing reviewed clips supply a small supervised pilot

Training item	Value
Feeding / Walking reviewed duration	60 s / 14 s
Derived two-second segments	30 Feeding + 7 Walking = 37
Motion representation	18 features, 96 × 96 grayscale full frames
Random Forest	200 trees, depth 8, balanced weights

Animal detection with MegaDetector

Pretrained YOLOv5 detector for animals, people and vehicles

The detector proposes an animal box and score.

It does not classify eating or sleep.

Control pass: 640-pixel input.
Animal threshold: 0.4.



Detector input 640



Detector input 1280

Same saved audit image. The red outline already exists in the input. Yellow shows the new proposal.

Image improvement experiment

Twelve saved audit stills, paired settings, identical detector weights

Variant / input size	Detected at 0.4	Detected at 0.2
raw / 640	1 / 12	3 / 12
raw / 1280	9 / 12	9 / 12
clahe / 1280	8 / 12	9 / 12
gamma_0.7 / 1280	9 / 12	9 / 12

Saved stills include ROI outlines. Extra proposals do not establish better accuracy.

Sampling and multi-camera fusion

One reproducible nominal two-second window in each five-minute stratum

Only 43.97 camera-minutes were sampled, about 0.67%.

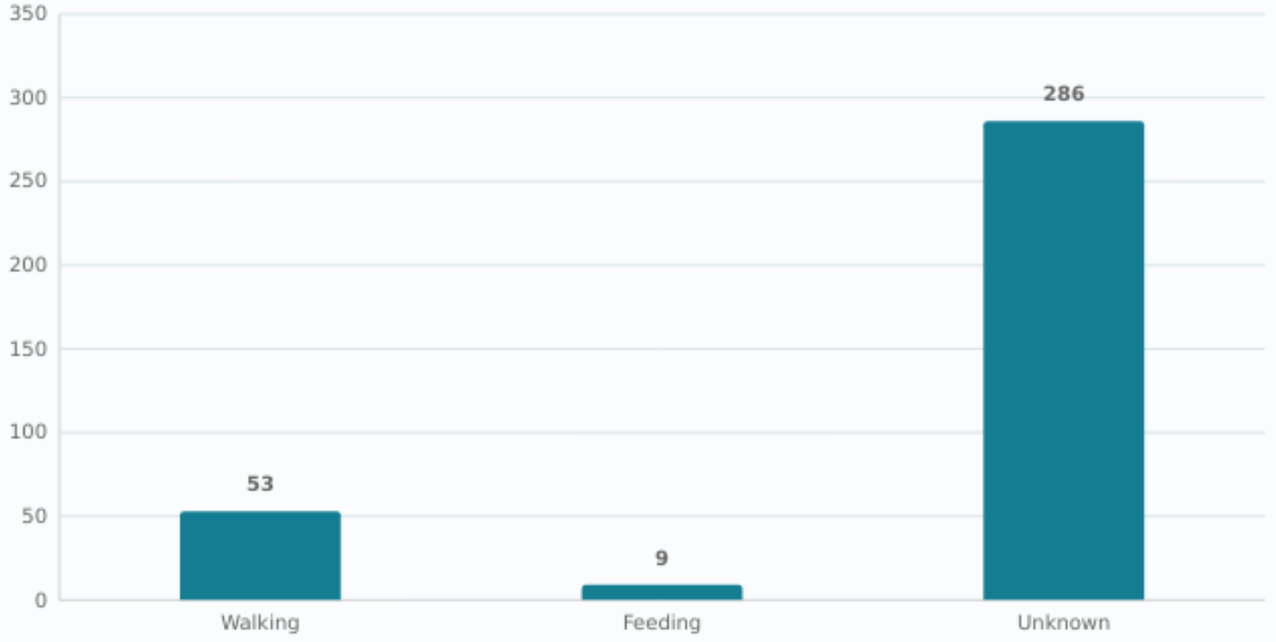
Combine matching nominal camera times once per animal.

Conflicting proposals become Unknown.

Do not fill unsampled gaps or turn Unknown into Inactive.

Main results from 348 sampled times

Historical run: model proposals with an explicit Unknown category

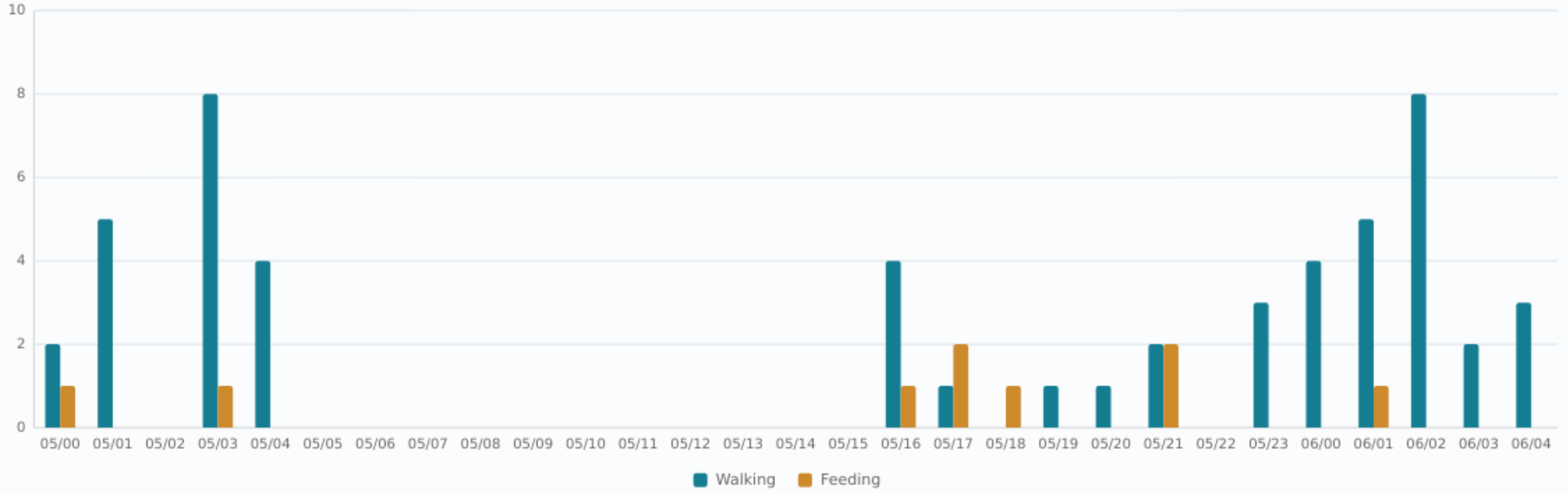


Walking	53	15.23%
Feeding	9	2.59%
Unknown	286	82.18%

Active proposals: 62 / 348
Verified Inactive: unavailable

Hourly sampled activity proposals

Each hourly bin contains 12 nominal sampled times



Peak: 9 Active proposals at 03:00 on 5 April. Zero proposals does not establish rest.

Additional automatic evidence

Saved audit clips and one additional original video form separate cohorts

77 / 77 windows completed
548 detector frames
10 frames with animal proposals
0 processing errors

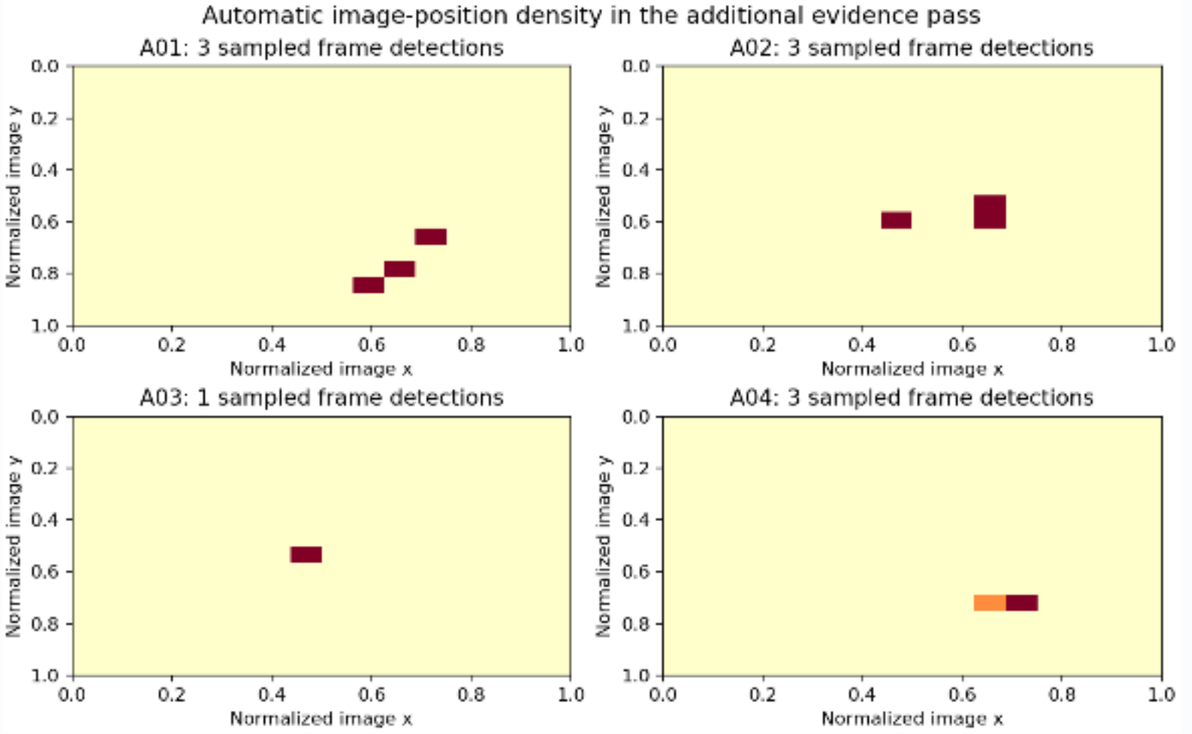


Image-coordinate density counts detected centroids. It does not measure room occupancy.

Evaluation and error analysis

Software correctness and behavior accuracy are separate checks

Historical table arithmetic and fusion: passed.

Software suite: 34 tests passed.

Walking development replay: 18 / 30 seconds proposed Walking.

Independent multi-class accuracy, precision, recall and F1: unavailable.

Reproducibility and fast trace

Open OPEN_THIS_FIRST.md in the GitHub repository

Training question: open model settings and annotation register.

Result question: open fused, per-video or hourly CSV.

Test question: open software checks or development replay evidence.

Method question: open source code and the automatic analysis report.

Conclusions

A working trained pilot with source-linked automatic results

The run produced Walking and Feeding proposals with Unknown.

Automation exports animal boxes, motion proxies and scoped plots.

Sampled results cannot establish a full-day activity budget.

Verified Inactive and independent behavior accuracy remain unavailable.